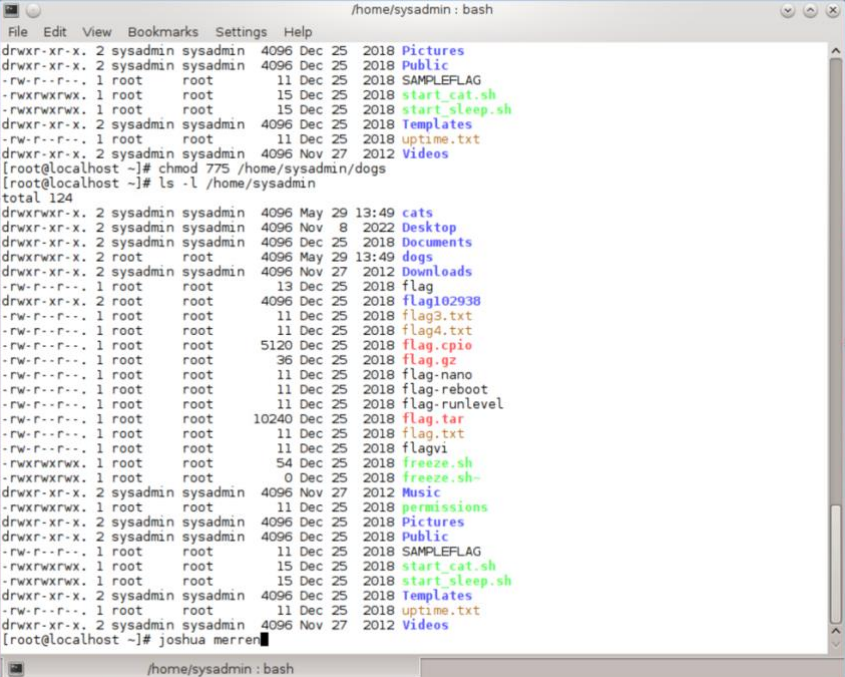
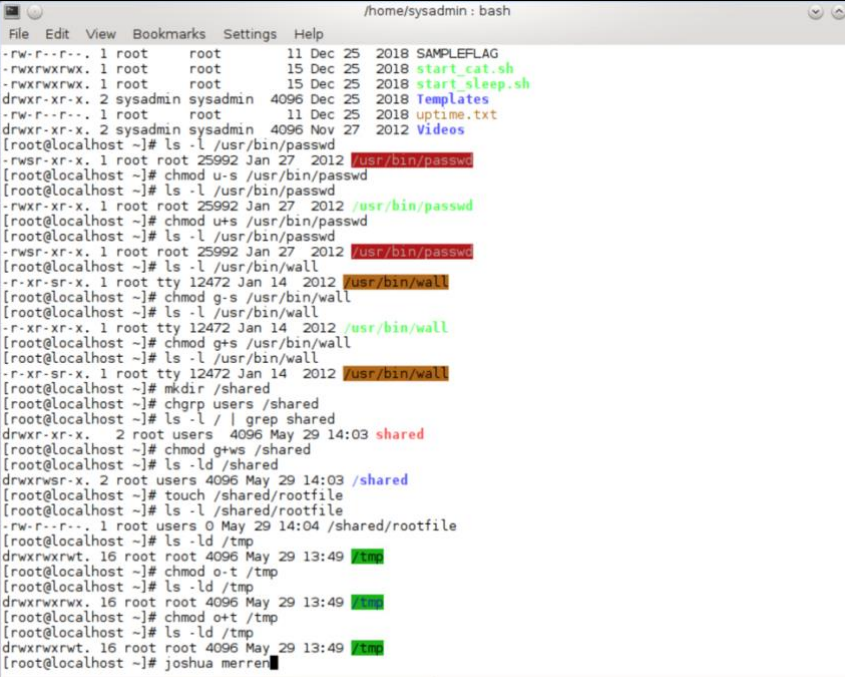


## CYB 230 Module Four Lab Worksheet

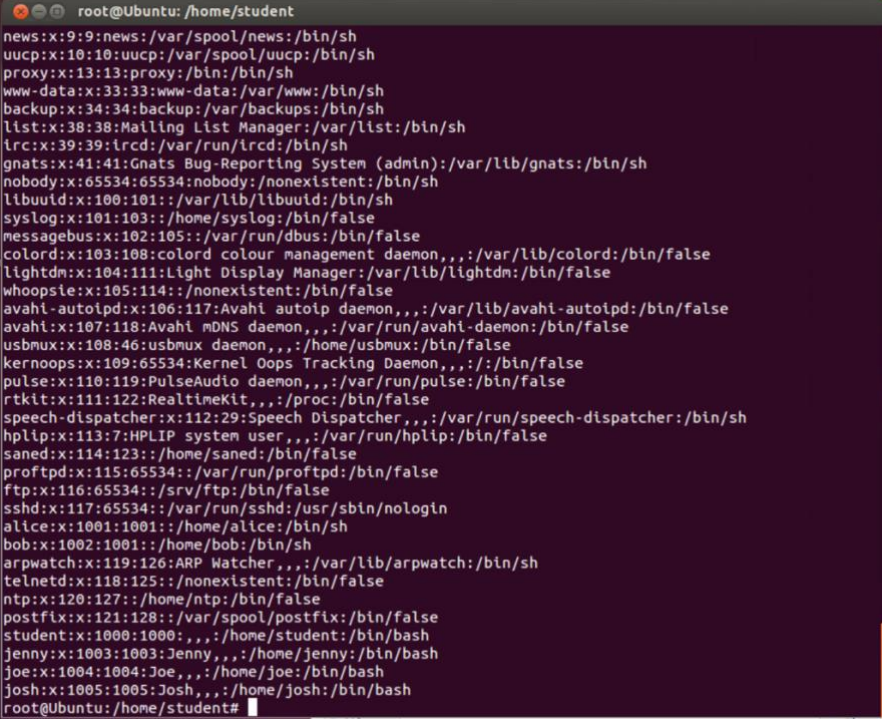
Complete this worksheet by replacing the bracketed phrases in the Response column with the relevant information.

### Lab: Working With Files

| Prompt                                                                                                                                                  | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| In the lab section "Using Chmod to Change Permissions," insert your name at the command line below the ending output and include it in your screenshot. |  <pre> /home/sysadmin: bash drwxr-xr-x. 2 sysadmin sysadmin 4096 Dec 25 2018 Pictures drwxr-xr-x. 2 sysadmin sysadmin 4096 Dec 25 2018 Public -rw-r--r--. 1 root root 11 Dec 25 2018 SAMPLEFLAG -rwxrwxrwx. 1 root root 15 Dec 25 2018 start_cat.sh -rwxrwxrwx. 1 root root 15 Dec 25 2018 start_sleep.sh drwxr-xr-x. 2 sysadmin sysadmin 4096 Dec 25 2018 Templates -rw-r--r--. 1 root root 11 Dec 25 2018 uptime.txt drwxr-xr-x. 2 sysadmin sysadmin 4096 Nov 27 2012 Videos [root@localhost ~]# chmod 775 /home/sysadmin/dogs [root@localhost ~]# ls -l /home/sysadmin total 124 drwxr-xr-x. 2 sysadmin sysadmin 4096 May 29 13:49 cats drwxr-xr-x. 2 sysadmin sysadmin 4096 Nov 8 2022 Desktop drwxr-xr-x. 2 sysadmin sysadmin 4096 Dec 25 2018 Documents drwxr-xr-x. 2 root root 4096 May 29 13:49 dogs drwxr-xr-x. 2 sysadmin sysadmin 4096 Nov 27 2012 Downloads -rw-r--r--. 1 root root 13 Dec 25 2018 flag drwxr-xr-x. 2 root root 4096 Dec 25 2018 flag102938 -rw-r--r--. 1 root root 11 Dec 25 2018 flag3.txt -rw-r--r--. 1 root root 11 Dec 25 2018 flag4.txt -rw-r--r--. 1 root root 5120 Dec 25 2018 flag.cpio -rw-r--r--. 1 root root 36 Dec 25 2018 flag.gz -rw-r--r--. 1 root root 11 Dec 25 2018 flag-nano -rw-r--r--. 1 root root 11 Dec 25 2018 flag-reboot -rw-r--r--. 1 root root 11 Dec 25 2018 flag-runlevel -rw-r--r--. 1 root root 10240 Dec 25 2018 flag.tar -rw-r--r--. 1 root root 11 Dec 25 2018 flag.txt -rw-r--r--. 1 root root 11 Dec 25 2018 flagvi -rwxrwxrwx. 1 root root 54 Dec 25 2018 freeze.sh -rwxrwxrwx. 1 root root 0 Dec 25 2018 freeze.sh- drwxr-xr-x. 2 sysadmin sysadmin 4096 Nov 27 2012 Music -rwxrwxrwx. 1 root root 11 Dec 25 2018 permissions drwxr-xr-x. 2 sysadmin sysadmin 4096 Dec 25 2018 Pictures drwxr-xr-x. 2 sysadmin sysadmin 4096 Dec 25 2018 Public -rw-r--r--. 1 root root 11 Dec 25 2018 SAMPLEFLAG -rwxrwxrwx. 1 root root 15 Dec 25 2018 start_cat.sh -rwxrwxrwx. 1 root root 15 Dec 25 2018 start_sleep.sh drwxr-xr-x. 2 sysadmin sysadmin 4096 Dec 25 2018 Templates -rw-r--r--. 1 root root 11 Dec 25 2018 uptime.txt drwxr-xr-x. 2 sysadmin sysadmin 4096 Nov 27 2012 Videos [root@localhost ~]# joshua merrin </pre> |
| In the lab section "Setting Special Permissions," insert your name at the command line below the ending output and include it in your screenshot.       |  <pre> /home/sysadmin: bash -rw-r--r--. 1 root root 11 Dec 25 2018 SAMPLEFLAG -rwxrwxrwx. 1 root root 15 Dec 25 2018 start_cat.sh -rwxrwxrwx. 1 root root 15 Dec 25 2018 start_sleep.sh drwxr-xr-x. 2 sysadmin sysadmin 4096 Dec 25 2018 Templates -rw-r--r--. 1 root root 11 Dec 25 2018 uptime.txt drwxr-xr-x. 2 sysadmin sysadmin 4096 Nov 27 2012 Videos [root@localhost ~]# ls -l /usr/bin/passwd -rwsr-xr-x. 1 root root 25992 Jan 27 2012 /usr/bin/passwd [root@localhost ~]# chmod u-s /usr/bin/passwd [root@localhost ~]# ls -l /usr/bin/passwd -rwsr-xr-x. 1 root root 25992 Jan 27 2012 /usr/bin/passwd [root@localhost ~]# chmod u+s /usr/bin/passwd [root@localhost ~]# ls -l /usr/bin/passwd -rwsr-xr-x. 1 root root 25992 Jan 27 2012 /usr/bin/passwd [root@localhost ~]# ls -l /usr/bin/wall -r-xr-sr-x. 1 root tty 12472 Jan 14 2012 /usr/bin/wall [root@localhost ~]# chmod g-s /usr/bin/wall [root@localhost ~]# ls -l /usr/bin/wall -r-xr-sr-x. 1 root tty 12472 Jan 14 2012 /usr/bin/wall [root@localhost ~]# chmod g+s /usr/bin/wall [root@localhost ~]# ls -l /usr/bin/wall -r-xr-sr-x. 1 root tty 12472 Jan 14 2012 /usr/bin/wall [root@localhost ~]# mkdir /shared [root@localhost ~]# chgrp users /shared [root@localhost ~]# ls -l /   grep shared drwxr-xr-x. 2 root users 4096 May 29 14:03 shared [root@localhost ~]# chmod g+ws /shared [root@localhost ~]# ls -ld /shared drwxrwsr-x. 2 root users 4096 May 29 14:03 /shared [root@localhost ~]# touch /shared/rootfile [root@localhost ~]# ls -l /shared/rootfile -rw-r--r--. 1 root users 0 May 29 14:04 /shared/rootfile [root@localhost ~]# ls -ld /tmp drwxrwxrwt. 16 root root 4096 May 29 13:49 /tmp [root@localhost ~]# chmod o-t /tmp [root@localhost ~]# ls -ld /tmp drwxrwxrwt. 16 root root 4096 May 29 13:49 /tmp [root@localhost ~]# chmod o+t /tmp [root@localhost ~]# ls -ld /tmp drwxrwxrwt. 16 root root 4096 May 29 13:49 /tmp [root@localhost ~]# joshua merrin </pre>                                                                                                                                                                                                                   |

| Prompt                                                                                                                                                                                                                                             | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementing the sticky bit on the directory can stop people from accidentally deleting files that they don't own. How can this technique be used to implement the concept of least privilege, and how can it be used to assure file availability? | Implementing the sticky bit on a directory is vital for enforcing the concept of least privilege and ensuring file availability. It restricts deletion permissions so users can only remove files they own, even if they have broader access to the directory. This prevents accidental or malicious deletions, helping maintain the integrity and availability of files in shared environments, thereby supporting seamless collaboration and data protection. |

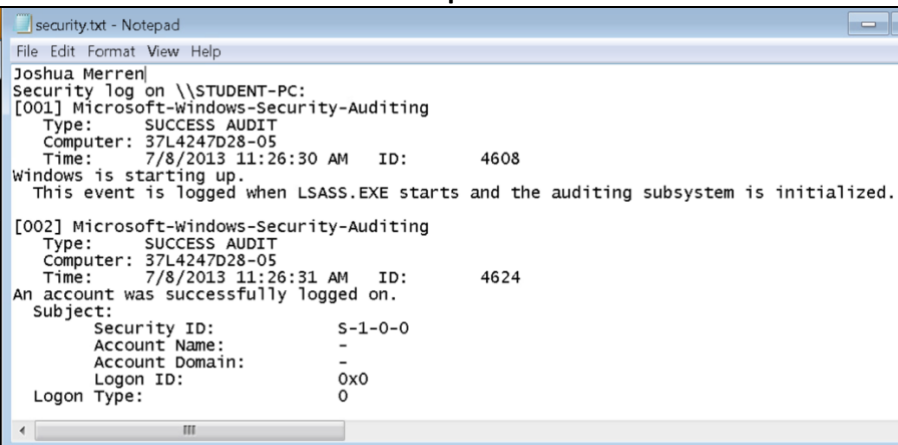
### Lab: Permissions, Users, and Groups in Linux

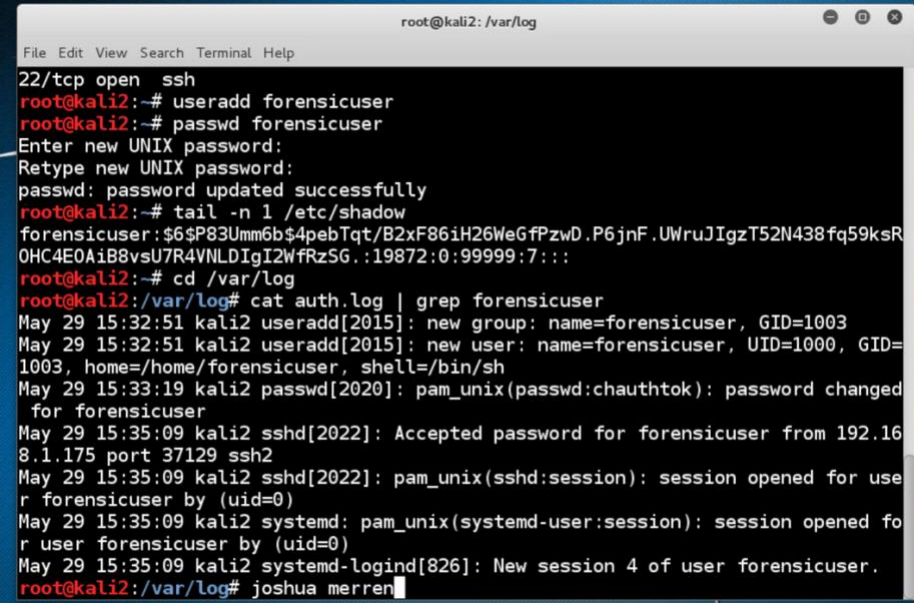
| Prompt                                                                                                                                                                                                                        | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| After completing the lab section "Adding Groups, Users, and Passwords," <b>repeat the steps to add another user using your first name. Provide a screenshot of the <code>cat etc/passwd</code> command when you are done.</b> |  <pre> root@Ubuntu: /home/student news:x:9:9:news:/var/spool/news:/bin/sh uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh proxy:x:13:13:proxy:/bin:/bin/sh www-data:x:33:33:www-data:/var/www:/bin/sh backup:x:34:34:backup:/var/backups:/bin/sh list:x:38:38:Mail List Manager:/var/list:/bin/sh irc:x:39:39:ircd:/var/run/ircd:/bin/sh gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh nobody:x:65534:65534:nobody:/nonexistent:/bin/sh libuuid:x:100:101::/var/lib/libuuid:/bin/sh syslog:x:101:103::/home/syslog:/bin/false messagebus:x:102:105::/var/run/dbus:/bin/false colord:x:103:108:colord colour management daemon,,:/var/lib/colord:/bin/false lightdm:x:104:111:Light Display Manager:/var/lib/lightdm:/bin/false whoopsie:x:105:114::/nonexistent:/bin/false avahi-autoipd:x:106:117:Avahi autoip daemon,,:/var/lib/avahi-autoipd:/bin/false avahi:x:107:118:Avahi mDNS daemon,,:/var/run/avahi-daemon:/bin/false usbmux:x:108:46:usbmux daemon,,:/home/usbmux:/bin/false kernoops:x:109:65534:Kernel Oops Tracking Daemon,,:/bin/false pulse:x:110:119:PulseAudio daemon,,:/var/run/pulse:/bin/false rtkit:x:111:122:RealtimeKit,,:/proc:/bin/false speech-dispatcher:x:112:29:Speech Dispatcher,,:/var/run/speech-dispatcher:/bin/sh hplip:x:113:7:HPLIP system user,,:/var/run/hplip:/bin/false saned:x:114:123::/home/saned:/bin/false proftpd:x:115:65534::/var/run/proftpd:/bin/false ftp:x:116:65534::/srv/ftp:/bin/false sshd:x:117:65534::/var/run/sshd:/usr/sbin/nologin alice:x:1001:1001::/home/alice:/bin/sh bob:x:1002:1001::/home/bob:/bin/sh arpwatch:x:119:126:ARP Watcher,,:/var/lib/arpwatch:/bin/sh telnetd:x:118:125::/nonexistent:/bin/false ntp:x:120:127::/home/ntp:/bin/false postfix:x:121:128::/var/spool/postfix:/bin/false student:x:1000:1000:::/home/student:/bin/bash jenny:x:1003:1003:Jenny,,:/home/jenny:/bin/bash joe:x:1004:1004:Joe,,:/home/joe:/bin/bash josh:x:1005:1005:Josh,,:/home/josh:/bin/bash root@Ubuntu: /home/student# </pre> |

| Prompt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <p>After completing the lab section “Absolute Permission,” repeat the process using your first name as the text file. Provide a screenshot of the output.</p> <p><b>Note:</b> By default, some computer systems use the key sequence <b>Ctrl+Alt+F1</b> to access a shortcut for other programs such as the Intel Graphics Control Panel. If this is the case, you will need to change the key sequence from the default to complete this step.</p> <p>To exit the tty1 or tty2 window, use the key sequence <b>Ctrl+Alt+F7</b>.</p> | <pre> Welcome to Ubuntu 12.04.5 LTS (GNU/Linux 3.13.0-32-generic i686)   * Documentation:  https://help.ubuntu.com/  2 packages can be updated. 0 updates are security updates.  New release '14.04.1 LTS' available. Run 'do-release-upgrade' to upgrade to it.  Your Hardware Enablement Stack (HWE) is supported until April 2017.  jenny@Ubuntu:~\$ cd /home jenny@Ubuntu:/home\$ ls -l total 20 drwxr-xr-x  3 jenny  jenny  4096 May 29 14:42 jenny drwx--r-x  3 joe    joe    4096 May 29 14:49 joe drwxr-xr-x  2 josh   josh   4096 May 29 14:25 josh drwxr-xr-x  2 root   root   4096 Apr 23  2015 scripts drwxr-xr-x 20 student student 4096 May 29 14:19 student jenny@Ubuntu:/home\$ cd /home/josh jenny@Ubuntu:/home/josh\$ ls -l total 12 -rw-r--r--  1 josh  josh 8445 May 29 14:25 examples.desktop jenny@Ubuntu:/home/josh\$ touch jenny.txt touch: cannot touch `jenny.txt': Permission denied jenny@Ubuntu:/home/josh\$                     </pre> |

| Prompt                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Response                                                                                                                              |
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| <p>Using the <b>chmod</b> command, which commands would you use to set the following permissions to a file called <b>Answers.txt</b>? (Provide the one line used at the command line for each bulleted item.)</p> <ul style="list-style-type: none"> <li>User (read and write), group (execute) other (execute)</li> <li>User (read, write, execute), group (read and execute) other (write and execute)</li> <li>User (write), group (read) other (none)</li> </ul> | <ul style="list-style-type: none"> <li>chmod 611 Answers.txt</li> <li>chmod 753 Answers.txt</li> <li>chmod 240 Answers.txt</li> </ul> |

### Lab: Log Analysis

| Prompt                                                                                                                                                         | Response                                                                             |
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| <p>In the lab section "Examining Windows Event Logs, IIS Logs, and Scheduled Tasks," add your name as the top line of the file and then take a screenshot.</p> |  |

| Prompt                                                                                                                                          | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| In the lab section "Examining Linux Log Files," insert your name at the command line below the ending output and include it in your screenshot. |  <pre> root@kali2: /var/log File Edit View Search Terminal Help 22/tcp open  ssh root@kali2:~# useradd forensicuser root@kali2:~# passwd forensicuser Enter new UNIX password: Retype new UNIX password: passwd: password updated successfully root@kali2:~# tail -n 1 /etc/shadow forensicuser:\$6\$P83Umm6b\$4pebTqt/B2xF86iH26WeGfPzwD.P6jnF.UWruJIgzT52N438fq59ksR OHC4E0AiB8vsU7R4VNLDIgI2WfRzSG.:19872:0:99999:7::: root@kali2:~# cd /var/log root@kali2:/var/log# cat auth.log   grep forensicuser May 29 15:32:51 kali2 useradd[2015]: new group: name=forensicuser, GID=1003 May 29 15:32:51 kali2 useradd[2015]: new user: name=forensicuser, UID=1000, GID= 1003, home=/home/forensicuser, shell=/bin/sh May 29 15:33:19 kali2 passwd[2020]: pam_unix(passwd:chauthtok): password changed for forensicuser May 29 15:35:09 kali2 sshd[2022]: Accepted password for forensicuser from 192.16 8.1.175 port 37129 ssh2 May 29 15:35:09 kali2 sshd[2022]: pam_unix(sshd:session): session opened for use r forensicuser by (uid=0) May 29 15:35:09 kali2 systemd: pam_unix(systemd-user:session): session opened fo r user forensicuser by (uid=0) May 29 15:35:09 kali2 systemd-logind[826]: New session 4 of user forensicuser. root@kali2:/var/log# joshua merren </pre> |
| What is the importance of maintaining clean log files that are well formatted?                                                                  | Maintaining clean and well-formatted log files is essential because it helps with efficient system troubleshooting and monitoring. Clean log files are easier to read and analyze, which speeds up the identification and resolution of issues. Well-formatted logs also support security measures by clearly showing all actions taken, making detecting unauthorized access or other malicious activities simpler. Additionally, clean logs are crucial for compliance with regulations that require accurate and accessible records of system activities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |